



TEAM THE DEVFOUNDERS

Ocean Trace · Maritime Oil-Spill Intelligence & Vessel Attribution

Sri Shakthi Institute of Engineering and Technology, Coimbatore · Smart India Hackathon 2026 · PS 26143

Ocean Trace

Research & References Compendium

SATELLITE SAR · DRIFT HINDCAST · AIS VESSEL ATTRIBUTION

PROBLEM STATEMENT 26143 · NATIONAL TECHNICAL RESEARCH ORGANISATION (NTRO)

Leveraging satellite imagery to determine oil spills at sea along with AIS data correlations to identify the vessel responsible for the spill.

Theme: Disaster Management · Category: Software · Smart India Hackathon 2026

PREPARED BY

A

Indhu Priyan M

Satellite SAR & Oil-Spill Detection

B

Mohan Kumar M

Oil-Spill Segmentation

C

Nandha Kumar J

Ocean Drift & Hindcast Modelling

D

Pavitra G V

Weather, Ocean & Environmental Data

E

Krishnan S A

AIS & Maritime Data Research

F

Keerthana M R

AIS Filtering & Vessel Attribution

Sri Shakthi Institute of Engineering and Technology, Coimbatore

Document	Research & References Compendium — Ocean Trace
Version / Date	2.0 · 28 September 2026
Prepared for	SIH 2026 evaluation panel · NTRO problem-statement owners
Status	Released for idea submission; to be revised after Grand Finale build

1. Executive Summary

Marine oil spills cause lasting damage to fisheries, coastlines and protected habitats, yet most spills detected from space are never attributed to a vessel. Problem Statement 26143 (NTRO) asks for an automated pipeline that detects and characterises slicks in satellite imagery, traces them back to an origin point and time, forecasts their spread, and ranks the vessels most likely responsible using historic AIS traffic.

Ocean Trace is our answer. This compendium is the evidence base behind it: every model, dataset, physical assumption, data source and legal reference used in the design is listed here with a plain statement of what we took from it and where it appears in the pipeline. The document is organised into six research tracks, each owned by one team member, so that responsibility for every design decision is traceable to a person and to a source.

In total the review covers 38 sources: 20 peer-reviewed papers, 4 labelled SAR datasets, 8 data services and APIs, 3 software frameworks and 3 legal or standards documents. It supports a two-stage detection design (YOLO11n → U-Net/ResNet-34), a hybrid physics + ML + Bayesian hindcast built on OpenDrift/OpenOil with ERA5 and CMEMS forcing, and an explainable six-factor AIS attribution score whose every factor is grounded in published maritime-surveillance research.



Earlier detection and a 72 h forecast let responders position booms before oil reaches the coast.



Reefs, mangroves and nesting beaches are the assets the pipeline is designed to protect.

This document at a glance

38 sources reviewed	6 research tracks / owners	10 pipeline stages covered
4 labelled SAR datasets compared	6 attribution factors, each sourced	3 legal frameworks for evidence

2. Team & Research Ownership

Ocean Trace is built as six research tracks that map one-to-one onto the stages of the pipeline. Each team member owns one track: they collected and read the sources listed under that section, decided what to adopt, and are responsible for that stage in the build. The letter code of each track is used as the reference-ID prefix throughout this document (e.g. C2 = second reference in Nandha Kumar's Drift & Hindcast track).

Code	Developer	Research ownership	Main focus
A	Indhu Priyan M SSIET, Coimbatore	Satellite SAR & Oil-Spill Detection	Sentinel-1, SAR pre-processing, YOLO11n detector, DARTIS dataset
B	Mohan Kumar M SSIET, Coimbatore	Oil-Spill Segmentation	U-Net, ResNet-34 encoder, Trujillo-Acatitla (Zenodo) dataset, polygon extraction
C	Nandha Kumar J SSIET, Coimbatore	Ocean Drift & Hindcast Modelling	OpenDrift/OpenOil physics, ML residual correction, Bayesian origin reconstruction, forecast
D	Pavitra G V SSIET, Coimbatore	Weather, Ocean & Environmental Data	ERA5, CMEMS, Open-Meteo, HYCOM, Stokes drift, environmental APIs
E	Krishnan S A SSIET, Coimbatore	AIS & Maritime Data Research	AIS sources & format (MarineCadastre, AISStream, DMA), synthetic AIS, data integration
F	Keerthana M R SSIET, Coimbatore	AIS Filtering & Vessel Attribution	Spatial / temporal / trajectory filtering, anomaly scoring, 6-factor ranking, evidence report

SSIET = Sri Shakthi Institute of Engineering and Technology, Coimbatore. All six members are B.E. students of the institute.

Pipeline and track mapping



A	Stages 1–3	Satellite data acquisition, SAR processing, YOLO11n detection
B	Stages 4–5	U-Net segmentation, polygon extraction, geometry & age
C	Stages 7–8	Backward hindcast, Bayesian origin, forward forecast, validation
D	Stage 6	ERA5 / CMEMS / Open-Meteo / HYCOM forcing fields
E	Stage 9	AIS ingest, cleaning, synthetic AIS, traffic reconstruction
F	Stages 9–10	Filtering, anomaly scoring, 6-factor ranking, evidence report

How to read each entry

Citation	Full bibliographic entry (authors, year, title, venue) so the source can be located and checked.
Summary	What the source actually establishes — in our own words, not the abstract.
What we take	The specific idea, number, dataset, algorithm or design decision that Ocean Trace adopted from it. This is the important part.
Used in	Which pipeline stage(s), slide or feasibility item the borrowed element appears in.

3. Literature Review Methodology

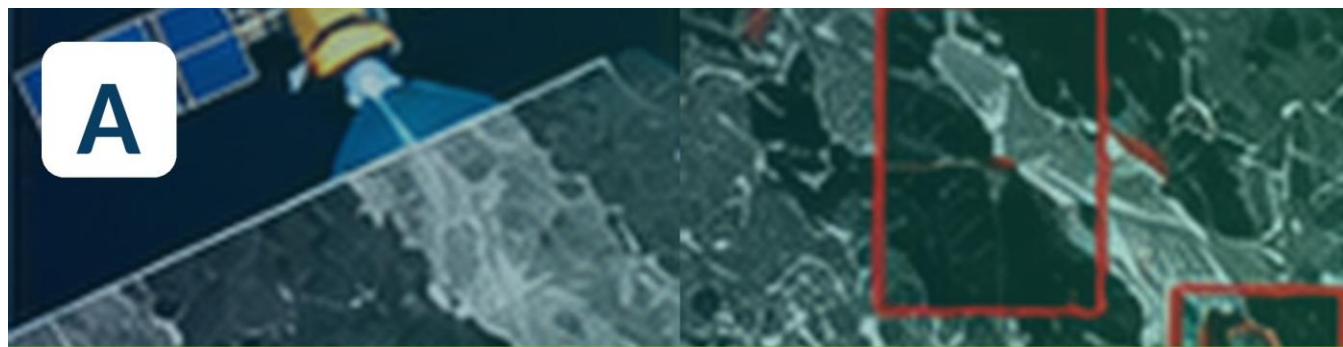
The review was carried out between August and September 2026 as part of preparing the SIH 2026 idea submission. Each owner followed the same five-step protocol so that the six tracks are comparable in depth and rigour.

1	Scope the question Each track owner wrote the two or three questions the pipeline stage must answer (e.g. 'Which polarisation and resolution give the best oil/look-alike separation in Sentinel-1?'). Sources were sought to answer those questions, not collected for volume.
2	Search Google Scholar, IEEE Xplore, MDPI, Copernicus journals (GMD, OS, ESSD), Zenodo and PANGAEA for datasets, and the official portals of ESA, ECMWF, Copernicus Marine, NOAA, ISRO and IMO for data and standards. Search terms combined the topic ('oil spill', 'SAR', 'look-alike', 'hindcast', 'AIS anomaly') with the method ('deep learning', 'Lagrangian', 'Bayesian').
3	Screen Inclusion: peer-reviewed or officially published; directly applicable to Sentinel-1 / C-band SAR, surface drift, or AIS analytics; openly accessible data where a dataset is cited. Exclusion: single-incident case studies with no reusable method; sources whose data cannot be obtained; pre-2005 papers unless they are the origin of a method still in use (e.g. Fay 1971).
4	Extract For every retained source the owner recorded a summary in their own words, the specific element adopted ('What we take'), and the stage where it is used. This is the four-part entry format used throughout Sections A–F.
5	Verify Bibliographic details were checked against the publisher, DOI or repository page. Dataset sizes and formats were confirmed from the repository record. Three sources present in an earlier draft could not be verified and were removed.

Quality and limitations

Limitations: the review favours open data and open-source tools because the hackathon build must be reproducible at zero cost; commercial SAR (e.g. ICEYE, Capella) and paid AIS providers were noted but not adopted. Indian-Ocean-specific spill datasets are scarce, so the detector is trained on Mediterranean and Gulf of Mexico data and will need regional fine-tuning at the Grand Finale stage.

4. Section A — Satellite SAR & Oil-Spill Detection



RESEARCH OWNER

Indhu Priyan M

Sri Shakthi Institute of Engineering and Technology, Coimbatore · Focus: Sentinel-1, SAR pre-processing, YOLO11n detector, DARTIS dataset

Why oil is visible in SAR at all, why look-alikes are the main failure mode, and which detector and dataset we chose for Stage-1 (YOLO11n on DARTIS).

A1 Journal paper · Ref. [1]

Brekke, C. & Solberg, A. H. S. (2005). Oil spill detection by satellite remote sensing. Remote Sensing of Environment, 95(1), 1–13.

Summary	Classic review of the physics of oil detection in SAR: oil damps the short capillary waves that give the sea its radar backscatter, so slicks appear as dark patches. Explains the wind-speed window (roughly 3–10 m/s) in which detection is reliable and catalogues look-alikes (low-wind areas, biogenic films, rain cells, ship wakes, current shear).
What we take	The physical basis for the whole detection stage; the wind-speed window and look-alike catalogue became our 'physics filter' (wind check + shape/contrast rules) that runs after the CNN.
Used in	Stage 2 SAR Processing, Stage 3 AI Detection, Feasibility risk #01

A2 Journal paper · Ref. [2]

Solberg, A. H. S. (2012). Remote sensing of ocean oil-spill pollution. Proceedings of the IEEE, 100(10), 2931–2945.

Summary	Surveys operational oil-spill monitoring services (e.g. CleanSeaNet) and the three-step pipeline used operationally: dark-spot detection → feature extraction → classification. Discusses confidence levels and the role of a human analyst.
What we take	The operational 'detect → characterise → analyst confirms' workflow. Our one-click analyst confirmation in the dashboard mirrors this; our confidence scores follow the same idea of graded certainty.
Used in	Stage 3–5, dashboard analyst confirmation

A3 Journal paper · Ref. [3]

Topouzelis, K. N. (2008). Oil spill detection by SAR images: dark formation detection, feature extraction and classification algorithms. Sensors, 8(10), 6642–6659.

Summary	Reviews dark-formation detection thresholds, the geometric/backscatter/contextual features that separate oil from look-alikes (area, perimeter, complexity, contrast, surrounding wind, distance to ships) and classifiers used on them.
What we take	The feature list for slick characterisation: area, perimeter, orientation, contrast and backscatter damping. These are exactly the geometric properties the PS asks us to compute in step (a).
Used in	Stage 5 Characterisation

A4 Journal paper · Ref. [4]

Fingas, M. & Brown, C. E. (2018). A review of oil spill remote sensing. Sensors, 18(1), 91.

Summary	Broad review across optical, IR, SAR and other sensors. Confirms SAR as the workhorse for wide-area, all-weather, day/night spill detection and notes limits on estimating slick thickness from SAR.
What we take	Justifies SAR-first design (Sentinel-1 / RISAT) with EO as optional confirmation; tells us not to over-claim thickness/volume from SAR alone, so we report area and age, not volume.
Used in	Data feasibility, Stage 1 acquisition

A5 Journal paper · Ref. [5]

Krestenitis, M., Orfanidis, G., Ioannidis, K., Avgerinakis, K., Vrochidis, S. & Kompatsiaris, I. (2019). Oil spill identification from satellite images using deep neural networks. Remote Sensing, 11(15), 1762.

Summary	Introduces a public Sentinel-1 oil-spill semantic-segmentation dataset (5 classes: sea, oil, look-alike, ship, land) and benchmarks U-Net, LinkNet, PSPNet and DeepLab variants. DeepLabv3+ and U-Net family models reach useful IoU on oil while look-alikes remain the hardest class.
What we take	Evidence that encoder-decoder CNNs work on Sentinel-1 oil masks, and that look-alikes must be an explicit class or filter. Motivated our two-stage design (detector first, segmenter second) instead of a single segmenter over whole scenes.
Used in	Stage 3 & 4 model selection

A6 Dataset + data paper · Ref. [6]

Yang, Y.-J., Singha, S., Goldman, R. & Schütte, F. (2025). Dataset of oil slicks, look-alikes and remarkable SAR signatures obtained from Sentinel-1 data in the Eastern Mediterranean Sea (DARTIS). Earth System Science Data, 17, 6807. <https://doi.org/10.5194/essd-17-6807-2025>

Summary	Labelled Sentinel-1 dataset from the Eastern Mediterranean (2019): 3,225 oil objects across 1,365 image patches plus 2,290 patches of look-alikes and other phenomena, with bounding-box style labels and an object-detector baseline. Look-alike patches are grouped by phenomenon.
What we take	Our Stage-1 training data. Bounding-box labels fit an object detector (YOLO11n) directly; the look-alike subgroups let us measure false alarms per phenomenon. Our reported mAP@0.5 = 0.626 and 3.1 % false alarms on clean sea are measured on this dataset.
Used in	Stage 3 AI Detection (YOLO11n), Technical feasibility

A7 Journal paper · Ref. [7]

Bianchi, F. M., Espeseth, M. M. & Borch, N. (2020). Large-scale detection and categorization of oil spills from SAR images with deep learning. Remote Sensing, 12(14), 2260 (arXiv:2006.13575).

Summary	Trains deep models on a very large KSAT-annotated Sentinel-1 archive (713 scenes, 2,093 spill events, 2014–2018) at 40 m resolution using the VV channel, and categorises spills by shape/type. Reports which polarisation and pre-processing work at scale.
What we take	Confirms VV polarisation and speckle-smoothed Sigma0 as the right input; shows tiled inference over full scenes is practical. Backs our 'tiled GPU inference on 10k × 10k scenes' strategy.
Used in	Stage 2 SAR Processing, Feasibility risk #06

A8 Software · Ref. [8]

Jocher, G. & Qiu, J. (2024). Ultralytics YOLO11 (software, version 11.0).
<https://github.com/ultralytics/ultralytics>

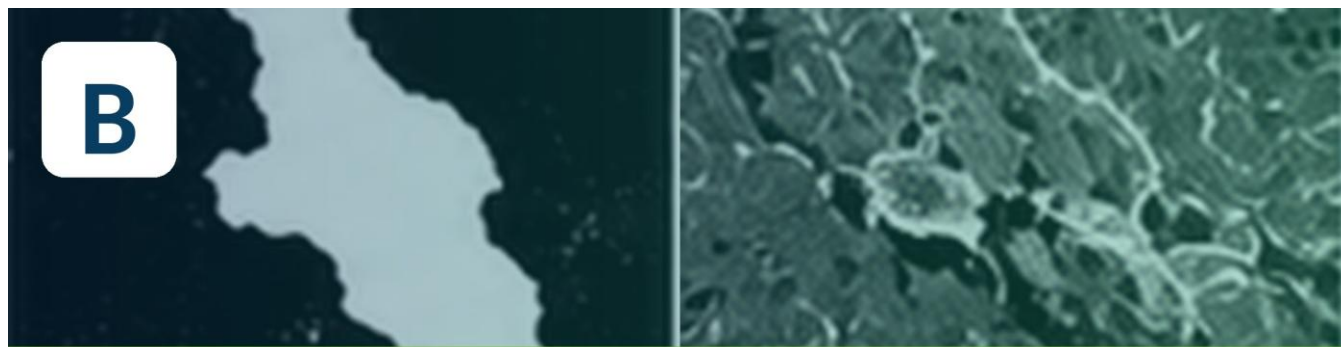
Summary	Latest Ultralytics detector family; the 'n' (nano) variant is designed for real-time inference on CPU/edge with ONNX/TensorRT export built in.
What we take	YOLO11n as the Stage-1 candidate detector: fast enough to screen every SAR pass, exportable to ONNX so the same weights run on CPU in the demo and on GPU in production.
Used in	Stage 3 AI Detection, Deployable AI

A9 Technical documentation · Ref. [9]

European Space Agency. Sentinel-1 SAR User Guide / Mission overview (Copernicus).
<https://sentinel.esa.int/web/sentinel/missions/sentinel-1>

Summary	Describes Sentinel-1 C-band SAR, the IW mode (250 km swath, ~10 m × 10 m GRD pixels), VV/VH polarisation, revisit (6–12 days per satellite) and the GRD product used for ocean applications.
What we take	Product choice (IW GRD, VV) and the revisit figures quoted in our risk table; the reason we also plan RISAT / EOS-04 as a second SAR source to shorten revisit gaps.
Used in	Stage 1 acquisition, Feasibility risk #06

5. Section B — Oil-Spill Segmentation



RESEARCH OWNER

Mohan Kumar M

Sri Shakthi Institute of Engineering and Technology, Coimbatore · Focus: U-Net, ResNet-34 encoder, Trujillo-Acatitla (Zenodo) dataset, polygon extraction

Turning a detected box into a pixel-accurate slick polygon (U-Net with ResNet-34 encoder on the Trujillo-Acatitla Zenodo dataset), then extracting geometry.

B1 Conference paper · Ref. [10]

Ronneberger, O., Fischer, P. & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. MICCAI 2015, LNCS 9351, 234–241.

Summary	The encoder–decoder architecture with skip connections that recovers fine spatial detail at the output. Works well with small datasets thanks to heavy augmentation.
What we take	Our Stage-2 segmenter backbone. Skip connections matter for thin, filament-like slick edges; augmentation strategy adopted because labelled SAR data is scarce.
Used in	Stage 4 Segmentation

B2 Conference paper · Ref. [11]

He, K., Zhang, X., Ren, S. & Sun, J. (2016). Deep residual learning for image recognition. IEEE CVPR 2016, 770–778.

Summary	Introduces residual (skip) blocks that let deep networks train stably; ResNet-34 is the lightweight member of the family.
What we take	ResNet-34 as the U-Net encoder: ImageNet-pretrained weights give a good start with little SAR data, and the model stays small enough for CPU inference via ONNX.
Used in	Stage 4 Segmentation, Deployable AI

B3 Dataset · Ref. [12]

Trujillo-Acatitla, R., Tuxpan-Vargas, J. et al. (2024). Sentinel-1 SAR oil spill image dataset for train, validate and test deep learning models, Parts I–II. Zenodo. <https://zenodo.org/records/8253899>

Summary	Sentinel-1 Sigma0 (dB) images at 2048×2048 pixels with pixel-level ground-truth masks in TIFF, split into oil, no-oil and look-alike sets (e.g. 685 no-oil and 685 look-alike training/validation scenes in Part II). This is the Zenodo dataset named in the problem statement.
What we take	Our Stage-2 training data for pixel-level masks. The no-oil and look-alike sets are used as hard negatives so the segmenter does not 'paint' oil on calm-sea patches.

Used in	Stage 4 Segmentation, Data feasibility
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B4 Journal paper · Ref. [13]

Shaban, M., Salim, R., Abu Khalifeh, H., Khelifi, A., Shalaby, A., El-Mashad, S., Mahmoud, A., Ghazal, M. & El-Baz, A. (2021). A deep-learning framework for the detection of oil spills from SAR data. *Sensors*, 21(7), 2351.

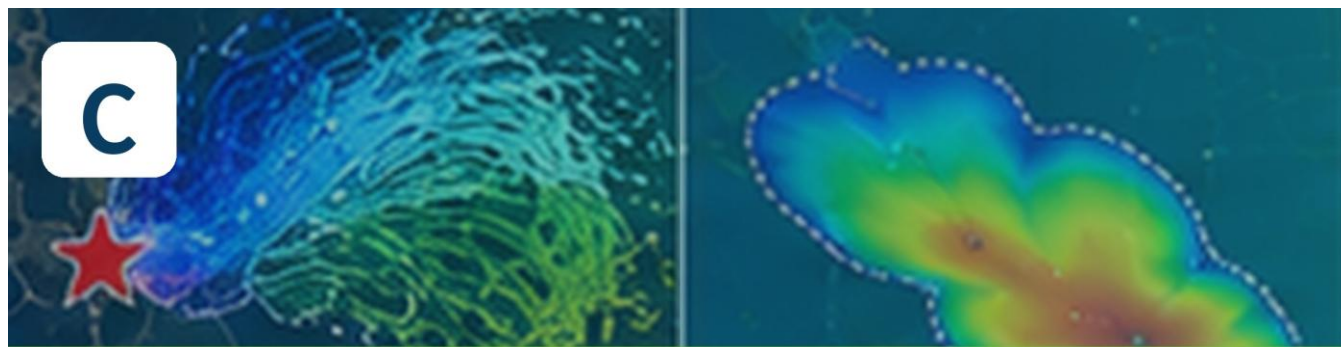
Summary	Two-stage framework: a CNN first classifies patches by oil content, then a U-Net segments only the positive patches; uses generalized Dice loss to handle the tiny fraction of oil pixels.
What we take	Direct precedent for our detector-then-segmenter pipeline and for using Dice-type loss on heavily imbalanced masks.
Used in	Stage 3–4 pipeline design, training loss

B5 Conference paper (classic) · Ref. [14]

Fay, J. A. (1971). Physical processes in the spread of oil on a water surface. *Proceedings of the Joint Conference on Prevention and Control of Oil Spills*, API, 463–467.

Summary	Describes the three spreading regimes (gravity–inertia, gravity–viscous, surface-tension–viscous) that govern how a slick's area grows with time.
What we take	The basis of our 'estimated age' figure: inverting the Fay spreading relations from observed area gives a rough age band (e.g. 1–3 days), which the PS asks for 'if feasible'. Reported as a range, never a point.
Used in	Stage 5 Characterisation (age)

6. Section C — Ocean Drift & Hindcast Modelling



RESEARCH OWNER

Nandha Kumar J

Sri Shakthi Institute of Engineering and Technology, Coimbatore · Focus: OpenDrift/OpenOil physics, ML residual correction, Bayesian origin reconstruction, forecast

Moving the slick backward to find the origin and time window, forward to predict spread, and validating the origin by re-simulation.

C1 Journal paper + open-source software · Ref. [15]

Dagestad, K.-F., Röhrs, J., Breivik, Ø. & Ådlandsvik, B. (2018). OpenDrift v1.0: a generic framework for trajectory modelling. Geoscientific Model Development, 11, 1405–1420.

Summary	Open-source Lagrangian particle framework: seeds particles, reads wind/current/wave fields from many formats, and integrates trajectories forward or backward in time with configurable diffusion.
What we take	Our drift engine. Native backward integration gives the hindcast; the same configuration run forward gives the forecast and the validation re-simulation. Multi-reader support lets us mix ERA5 + CMEMS.
Used in	Stage 7 Hindcast, Stage 8 Forecast, Bidirectional verification

C2 Journal paper · Ref. [16]

Röhrs, J., Dagestad, K.-F., Asbjørnsen, H., Nordam, T., Skancke, J., Jones, C. E. & Brekke, C. (2018). The effect of vertical mixing on the horizontal drift of oil spills. Ocean Science, 14, 1581–1601.

Summary	Presents OpenOil, the oil module of OpenDrift: wave entrainment, resurfacing and droplet size affect how much of the slick 'feels' the wind versus the sub-surface current, changing drift speed and direction.
What we take	Why a pure surface-current model is wrong for oil; the OpenOil weathering/entrainment settings we use, and the physical argument for running an ensemble (500 particles) rather than one trajectory.
Used in	Stage 7–8, Feasibility risk #03

C3 Journal paper · Ref. [17]

Lehr, W., Jones, R., Evans, M., Simecek-Beatty, D. & Overstreet, R. (2002). Revisions of the ADIOS oil spill model. Environmental Modelling & Software, 17(2), 189–197.

Summary	NOAA's oil weathering model: evaporation, emulsification, dispersion and density/viscosity change as a function of oil type, wind and time.
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What we take	The weathering terms in our forecast (mass loss and viscosity growth over 24/48/72 h) and part of the age estimate cross-check.
Used in	Stage 8 Forecast (weathering model)

C4 Journal paper · Ref. [18]

Breivik, Ø., Allen, A. A., Maisondieu, C. & Olagnon, M. (2013). Advances in search and rescue at sea. Ocean Dynamics, 63, 83–88.

Summary	Explains ensemble/stochastic drift prediction and probability maps used in SAR (search-and-rescue): many perturbed particles → probability of presence rather than a single point.
What we take	The template for our probabilistic outputs: P50 / P90 confidence hulls for both origin and future footprint instead of a single point or path.
Used in	Stage 7–8 uncertainty regions

C5 Journal paper · Ref. [19]

Breivik, Ø. & Allen, A. A. (2008). An operational search and rescue model for the Norwegian Sea and the North Sea. Journal of Marine Systems, 69(1–2), 99–113.

Summary	Operational leeway/drift model combining wind drag, current and Stokes drift, with an explicit treatment of the errors in each forcing field.
What we take	How to add Stokes drift and a wind-drift factor on top of currents, and the idea of a residual/error term per forcing — the seed of our ML residual-correction model that learns from past spill tracks.
Used in	Stage 7 ML residual correction

C6 Textbook · Ref. [20]

Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer. (Chapters on Bayesian inference and mixture models.)

Summary	Standard reference for Bayesian estimation: prior × likelihood → posterior, maximum-a-posteriori (MAP) estimates and credible regions.
What we take	The formal basis of 'Bayesian fusion': the backward particle cloud is the likelihood over origin position/time, AIS traffic density acts as a prior, and we report the MAP origin with 50 % / 90 % credible regions.
Used in	Stage 7 Bayesian fusion, Probabilistic Origin Intelligence

7. Section D — Weather, Ocean & Environmental Data



RESEARCH OWNER

Pavitra G V

Sri Shakthi Institute of Engineering and Technology, Coimbatore · Focus: ERA5, CMEMS, Open-Meteo, HYCOM, Stokes drift, environmental APIs

The forcing fields the drift model needs — wind, currents, waves — and where to get them free, historically and in near-real time.

D1 Journal paper + dataset · Ref. [21]

Hersbach, H. et al. (2020). The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society, 146(730), 1999–2049.

Summary	ECMWF's fifth-generation reanalysis: hourly global atmospheric fields (10 m wind, waves, etc.) from 1940 onward at ~31 km, freely available through the Copernicus Climate Data Store.
What we take	Our primary historical wind source for backward hindcasts (spills are usually seen days after release, so reanalysis is essential). Hourly resolution matches our 10-minute particle steps after interpolation.
Used in	Stage 6 Environmental Data, Stage 7 Hindcast

D2 Dataset / API · Ref. [22]

Copernicus Marine Service (CMEMS). Global Ocean Physics Analysis and Forecast / Reanalysis products (currents, temperature, waves). <https://marine.copernicus.eu>

Summary	Free global and regional ocean products: surface and sub-surface currents, sea-surface temperature, wave height and Stokes drift, both reanalysis (for hindcast) and 10-day forecast (for prediction).
What we take	Currents and Stokes-drift fields for OpenOil; the forecast product drives our 24/48/72 h prediction. Indian Ocean coverage confirmed for our region.
Used in	Stage 6, Stage 7–8

D3 API · Ref. [23]

Open-Meteo. Free weather and marine API (historical and forecast). <https://open-meteo.com>

Summary	Open, key-free HTTP API exposing ERA5, GFS and marine (wave) variables for any point, with historical archive access.
What we take	Lightweight fallback and demo source when CDS/CMEMS downloads are slow; keeps the hackathon prototype working without large downloads.

Used in	Stage 6 multi-provider fallback
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D4 Dataset / model · Ref. [24]

NOAA / NCEP. Global Forecast System (GFS) and HYCOM Global Ocean Forecast System.

Summary	US operational global atmosphere (GFS) and ocean (HYCOM) forecast systems, freely distributed.
What we take	Second independent provider for wind and currents, so an outage or licence issue on the Copernicus side does not stop the pipeline; also useful to estimate forcing-field uncertainty by comparing providers.
Used in	Stage 6 multi-provider fallback

D5 Data portal / API · Ref. [25]

Copernicus Data Space Ecosystem (CDSE) and Alaska Satellite Facility (ASF). Sentinel-1 GRD scene search and download. <https://dataspace.copernicus.eu> · <https://search.asf.alaska.edu>

Summary	Two free portals with APIs for searching Sentinel-1 scenes by area, date and mode and downloading GRD products.
What we take	Automated scene acquisition in Stage 1; ASF as a mirror when CDSE quotas bite.
Used in	Stage 1 Satellite Data Acquisition

D6 Mission documentation · Ref. [26]

ISRO. EOS-04 (RISAT-1A) C-band SAR mission overview. <https://www.isro.gov.in>

Summary	Indian C-band SAR satellite with medium and high-resolution modes suited to ocean surveillance.
What we take	Planned Indian data source to cut revisit time over the Indian EEZ; since it is also C-band, the same detector design applies with re-calibration.
Used in	Feasibility risk #06, Roadmap

8. Section E — AIS & Maritime Data Research



RESEARCH OWNER

Krishnan S A

Sri Shakthi Institute of Engineering and Technology, Coimbatore · Focus: AIS sources & format (MarineCadastre, AISStream, DMA), synthetic AIS, data integration

What AIS is, what a message contains, where historic AIS can be obtained, and how to build synthetic AIS for the spill region when real data is unavailable.

E1 Dataset + format specification · Ref. [27]

NOAA Office for Coastal Management / BOEM. MarineCadastre.gov AIS data and format specification (AccessAIS). <https://marinecadastre.gov/ais>

Summary	Daily CSV extracts of US-coast AIS with the standard fields: MMSI, timestamp, LAT/LON, SOG, COG, heading, vessel name, IMO, call sign, vessel type, status, length, width, draft, cargo.
What we take	The schema for our AIS ingest and for the synthetic AIS generator (the PS names this format explicitly). Every AIS source we use is normalised to these columns.
Used in	Stage 9 AIS Intelligence, Data feasibility

E2 API · Ref. [28]

AISStream.io. Real-time global AIS websocket API. <https://aisstream.io>

Summary	Free websocket feed of live terrestrial AIS messages worldwide, decoded to JSON.
What we take	Live vessel positions for the dashboard and for building a small historical archive over the demo region during the hackathon.
Used in	Stage 9, Dashboard

E3 Dataset · Ref. [29]

Danish Maritime Authority (DMA). Historical AIS data (open). <https://web.ais.dk/aisdata/>

Summary	Large open archive of decoded historical AIS CSVs (Danish waters), widely used in AIS research.
What we take	Realistic historic traffic to test the filtering and ranking algorithms at scale before applying them to the (sparser) Indian-region data.
Used in	Stage 9–10 testing

E4 Journal paper (survey) · Ref. [30]

Tu, E., Zhang, G., Rachmawati, L., Rajabally, E. & Huang, G.-B. (2018). Exploiting AIS data for intelligent maritime navigation: A comprehensive survey from data to methodology. IEEE Transactions on Intelligent Transportation Systems, 19(5), 1559–1582.

Summary	Survey of AIS data quality problems (gaps, spoofing, duplicate MMSIs, position jumps) and of methods for trajectory cleaning, reconstruction, prediction and anomaly detection.
What we take	Our AIS cleaning steps (deduplication, speed-jump rejection, gap interpolation limits) and the vocabulary for 'gaps & noise' in the challenges. Also the survey of trajectory-similarity measures used in Section F.
Used in	Stage 9 AIS cleaning, Challenge 'AIS gaps & noise'

E5 Journal paper · Ref. [31]

Kroodsma, D. A. et al. (2018). Tracking the global footprint of fisheries. Science, 359(6378), 904–908.

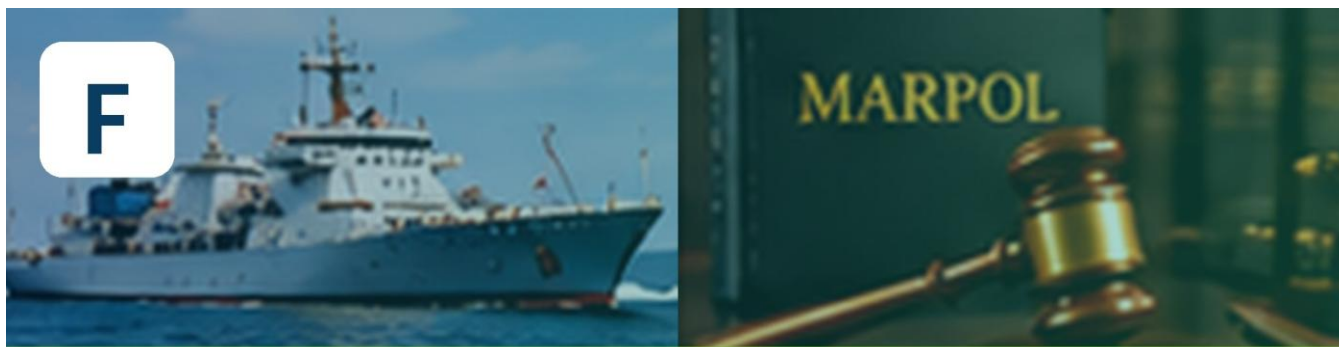
Summary	Global Fishing Watch: applies neural networks to AIS tracks to infer vessel behaviour (fishing vs transit) from speed/course patterns at global scale.
What we take	Proof that behaviour can be inferred from AIS kinematics alone; template for our behavioural features (speed drops, loitering, course changes) and for handling billions of messages efficiently.
Used in	Stage 10 behavioural anomaly features

E6 Standard / regulation · Ref. [32]

International Maritime Organization. SOLAS Chapter V, Regulation 19 (AIS carriage requirement) and ITU-R M.1371 (AIS technical standard).

Summary	Defines which vessels must carry AIS (≥ 300 GT international, all passenger ships) and the message types/fields transmitted.
What we take	Why small or non-compliant vessels are 'dark' and why an AIS blackout near the origin is itself suspicious; sets expectations for which suspects can even appear in AIS.
Used in	Challenge 'dark vessels', Stage 10 AIS-gap score

9. Section F — AIS Filtering & Vessel Attribution



RESEARCH OWNER

Keerthana M R

Sri Shakthi Institute of Engineering and Technology, Coimbatore · Focus: Spatial / temporal / trajectory filtering, anomaly scoring, 6-factor ranking, evidence report

From thousands of tracks to a short, explainable list of suspects: spatial → temporal → trajectory filtering, anomaly scoring, and the 6-factor ranking.

F1 Journal paper · Ref. [33]

Pallotta, G., Vespe, M. & Bryan, K. (2013). Vessel pattern knowledge discovery from AIS data: A framework for anomaly detection and route prediction. *Entropy*, 15(6), 2218–2245.

Summary	TREAD framework: unsupervised extraction of normal routes and waypoints from AIS, then flagging deviations (off-route, unusual speed, unexpected stops) as anomalies.
What we take	Our definition of 'anomalous behaviour' (factor 4): deviation from learned normal routes near the origin window. Route-prediction ideas support the trajectory-similarity factor.
Used in	Stage 10 factors 3–4

F2 Journal paper (review) · Ref. [34]

Riveiro, M., Pallotta, G. & Vespe, M. (2018). Maritime anomaly detection: A review. *WIREs Data Mining and Knowledge Discovery*, 8(5), e1266.

Summary	Categorises maritime anomalies (positional, kinematic, contextual, AIS-switch-off) and reviews statistical and ML detectors, stressing explainability for operators.
What we take	The taxonomy behind our anomaly features and the requirement that every score be explainable — hence a weighted, transparent 6-factor sum rather than a black-box classifier.
Used in	Stage 10 explainable ranking

F3 Journal paper · Ref. [35]

Ferraro, G., Meyer-Roux, S., Muellenhoff, O., Pavliha, M., Svetak, J., Tarchi, D. & Topouzelis, K. (2009). Long term monitoring of oil spills in European seas. *International Journal of Remote Sensing*, 30(3), 627–645.

Summary	JRC study correlating thousands of SAR-detected slicks with shipping lanes; shows most operational (deliberate) discharges lie along traffic routes and that spill–traffic correlation is the practical route to attribution.
What we take	Justifies the AIS-traffic prior in Bayesian fusion and the 'proximity to spill trajectory' factor; also the 'polluter pays' impact framing.

Used in	Stage 7 prior, Stage 10 factor 1, Impact slide
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F4 Journal paper · Ref. [36]

Vespe, M., Greidanus, H. & Alvarez, M. A. (2015). The declining impact of piracy on maritime transport in the Indian Ocean: statistical analysis of 5-year vessel tracking data. Marine Policy, 59, 9–15.

Summary	Uses satellite AIS over the Indian Ocean to reconstruct traffic density and route changes over time.
What we take	Confirms that historic AIS reconstruction of Indian-Ocean traffic at the scale we need is feasible, and gives a picture of typical route corridors for our synthetic-AIS generator.
Used in	Stage 9 traffic reconstruction, synthetic AIS

F5 Convention / regulation · Ref. [37]

International Maritime Organization. MARPOL 73/78, Annex I – Regulations for the prevention of pollution by oil.

Summary	Sets discharge limits (e.g. 15 ppm oily water, distance-from-land rules) and record-keeping (Oil Record Book) that a polluting vessel would have violated.
What we take	What an evidence pack must show for enforcement: time, position, vessel identity and the discharge inference. Shapes the fields of our generated investigation report.
Used in	Stage 10 Evidence generation, Actionable output

F6 National legislation · Ref. [38]

Government of India. Merchant Shipping Act, 1958 (Part XI-A, Prevention and containment of pollution of the sea by oil) and the Merchant Shipping (Prevention of Pollution of the Sea by Oil) Rules.

Summary	Domestic legal basis for penalising oil pollution from ships in Indian waters and for Coast Guard / DG Shipping action.
What we take	The Indian enforcement chain our report is written for (NTRO → ICG → DG Shipping / courts); ensures the evidence format matches what these bodies need.
Used in	Stakeholders, Actionable output

10. Labelled SAR Datasets — Comparison

Four public Sentinel-1 oil-spill datasets were evaluated. Two are used for training (DARTIS for the detector, Trujillo-Acatitla for the segmenter); the other two inform design and benchmarking.

Dataset	Sensor / region	Size	Label type	Role in Ocean Trace	Ref
DARTIS (Yang et al., 2025)	Sentinel-1, E. Mediterranean 2019	3,225 oil objects / 1,365 patches + 2,290 look-alike patches	Bounding boxes + look-alike subgroups	Stage-1 detector (YOLO11n) training and false-alarm testing	A6
Trujillo-Acatitla et al. (2024), Zenodo	Sentinel-1 Sigma0 (dB), 2048 × 2048 TIFF	Oil, no-oil and look-alike sets (e.g. 685 + 685 train/val scenes in Part II)	Pixel masks	Stage-2 segmenter (U-Net) training; hard negatives	B3
Krestenitis et al. (2019)	Sentinel-1, mixed regions	~1,000 images, 1250 × 650 px	5-class pixel masks (sea, oil, look-alike, ship, land)	Benchmarking segmenter architectures; ship class useful for AIS cross-check	A5
KSAT archive (Bianchi et al., 2020)	Sentinel-1 VV, global, 2014–2018	713 scenes, 2,093 spill events	Binary masks (not fully public)	Design guidance only: polarisation, resolution, tiling	A7

Research gaps and how Ocean Trace addresses them

Gap in the literature	What we observed	Ocean Trace contribution
Detection stops at the slick	Most published work ends at a segmented mask; origin and responsibility are left to analysts.	Ocean Trace chains detection → hindcast → attribution in one automated run with an analyst confirmation step.
Single-trajectory hindcasts	Many drift studies show one back-trajectory, which over-states certainty.	500-particle ensemble with Bayesian fusion; origin reported as P50 / P90 regions and a time window, then validated by forward re-simulation.
AIS matched by nearest vessel	Attribution is often 'closest ship at the time', which fails under dense traffic and AIS gaps.	Six-factor explainable score (proximity, timing, trajectory, behaviour, AIS gaps, vessel type) with per-factor evidence.
No operational zoning	Investigations treat the whole search area as one task, slowing multi-officer response.	Dynamic Maritime Zone Splitting: the origin hull is divided into zones, each with a bounded AIS search, worked in parallel and merged.
Regional data scarcity	Public spill datasets are Mediterranean / Gulf of Mexico; Indian EEZ examples are rare.	Transfer-learn from public sets, plan EOS-04 (RISAT-1A) ingestion and a regional fine-tuning set with NTRO / ICG at the Finale stage.

11. Design Decision Log

Every major architectural choice, the alternatives considered, the reason for the choice and the references that support it. This table is the bridge between the literature and the Technical Approach slide.

Decision	Alternatives considered	Why	Evidence
Two-stage detector + segmenter instead of one whole-scene segmenter	Single U-Net over full scenes; classical dark-spot thresholding	Detector screens $10k \times 10k$ scenes fast; segmenter only runs on candidates, cutting compute and false positives	A5 , A7 , B4
YOLO11n (nano) rather than a larger detector	YOLO11m/l, Faster R-CNN	Real-time on CPU for the demo; ONNX export; accuracy loss acceptable because Stage-2 refines the mask	A8 , A6
ResNet-34 encoder for U-Net	ResNet-50/101, EfficientNet	Pretrained weights, small enough for CPU inference, proven on SAR masks	B1 , B2
OpenDrift / OpenOil as drift engine	Custom particle code; GNOME	Open source, native backward integration, oil weathering module, multi-reader forcing	C1 , C2
ML residual correction on top of physics	Pure physics; pure data-driven drift	Physics gives structure; residual model absorbs forcing-field bias learned from past spill tracks	C5
Bayesian fusion for origin	Take the ensemble centroid	Produces MAP + credible regions and lets AIS traffic act as a prior	C4 , C6 , F3
ERA5 + CMEMS as primary forcing, Open-Meteo / GFS / HYCOM as fallback	Single provider	Availability and licence resilience; provider spread estimates forcing uncertainty	D1–D4
MarineCadastre schema for all AIS	Provider-specific formats	Named in the PS; one canonical schema simplifies synthetic AIS generation	E1
Weighted explainable score, not a black-box classifier	Gradient-boosted classifier on vessel features	Operators and courts need per-factor reasons; no labelled culprit dataset exists to train a classifier	F1 , F2

12. Traceability Matrix — Pipeline Stage → References

Reverse index: for every stage of the Ocean Trace pipeline (as drawn on the Technical Approach slide), the references that justify its design. Use this when an evaluator asks “where did this come from?”.

Pipeline stage / feature	Supporting references
1. Satellite Data Acquisition	A4, A9, D5
2. SAR Processing (calibration, speckle, masking)	A1, A7, A9
3. AI Detection (YOLO11n)	A1, A2, A5, A6, A8, B4
4. Segmentation (U-Net + ResNet-34)	A5, B1, B2, B3, B4
5. Characterisation (geometry, age)	A3, B5, C3
6. Environmental Data (wind, currents, waves)	D1, D2, D3, D4
7. Hindcast — origin estimation & Bayesian fusion	C1, C2, C4, C5, C6, F3
8. Forecast — spread prediction & weathering	C1, C2, C3, C4, D2
Bidirectional verification (forward re-sim vs slick)	C1, C4
9. AIS Intelligence (ingest, cleaning, synthetic AIS)	E1, E2, E3, E4, E6, F4
Dynamic Maritime Zone Splitting	C4, F1, F2
10. Attribution — 6-factor ranking & evidence	E5, F1, F2, F3, F5, F6
Deployment (ONNX, CPU/GPU, tiled inference)	A6, A8, B2

Six-factor vessel score — source of each factor

#	Factor	Derived from
1	Proximity to spill trajectory	F3 (spill–traffic correlation), C4 (probability hull)
2	Temporal compatibility with release window	C1, C6 (posterior time window)
3	Trajectory similarity	E4 (similarity measures), F1 (route knowledge)
4	Anomalous behaviour	F1, F2, E5
5	AIS gap / blackout analysis	E4, E6, F2
6	Vessel-type prior	F3, F5 (tankers & MARPOL Annex I)

13. Glossary

AIS	Automatic Identification System — VHF broadcast of a vessel's identity, position, course and speed.
Backscatter (Sigma0)	The radar energy returned to the SAR sensor; oil damps capillary waves and lowers backscatter, appearing dark.
Bayesian fusion	Combining a prior belief (e.g. traffic density) with evidence (the backward particle cloud) to obtain a posterior over origin and time.
CMEMS	Copernicus Marine Environment Monitoring Service — free ocean current, wave and temperature products.
DARTIS	Dataset of oil slicks, look-alikes and remarkable SAR signatures (Yang et al., 2025), used for detector training.
ERA5	ECMWF's fifth-generation atmospheric reanalysis — hourly wind fields used for hindcasting.
GRD	Ground Range Detected — the Sentinel-1 product type used for ocean applications.
Hindcast	Running the drift model backward in time from the observed slick to estimate where and when it was released.
Look-alike	A dark SAR feature that is not oil: low wind, biogenic film, rain cell, ship wake, current shear.
MAP	Maximum a posteriori — the single most probable origin under the Bayesian posterior.
MARPOL Annex I	IMO convention chapter regulating oil discharges from ships.
ONNX	Open Neural Network Exchange — portable model format enabling CPU/GPU inference without PyTorch.
P50 / P90	Regions containing 50 % / 90 % of the probability mass of origin or future spill position.
SAR	Synthetic Aperture Radar — all-weather, day/night imaging radar (Sentinel-1, RISAT/EOS-04).
Stokes drift	Net surface transport caused by waves, added to current and wind drift for floating oil.
VV polarisation	Vertical transmit / vertical receive radar channel — preferred for oil detection due to lower noise.

14. Consolidated Reference List

Numbered in order of appearance. The bracketed number is the citation number; the letter code is the research track and owner.

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Verification note: journal, conference and dataset entries were checked against their publisher or repository pages during preparation. Web resources (APIs, portals, mission pages) were accessed September 2026.